

Omics	Model	Spearman rho with age	AUC	AUC age only	AUC difference	P (BH)
Metabolomics	EN omics	+0.20	0.722	0.577	+0.146	0.143
Metabolomics	RF clinical (reference)	+0.43	-	-	-	-
Metabolomics	XGB omics	+0.27	0.737	0.577	+0.160	0.143
Microbiome	EN omics	+0.04	0.620	0.691	-0.071	0.262
Microbiome	RF clinical (reference)	+0.67	-	-	-	-
Microbiome	XGB omics	-0.04	0.586	0.691	-0.105	0.143
Proteomics	EN omics	+0.36	0.695	0.573	+0.121	0.143
Proteomics	RF clinical (reference)	+0.50	-	-	-	-
Proteomics	XGB omics	+0.41	0.674	0.573	+0.101	0.143